

PROJECT DESIGN DOCUMENT

MLE Proof-of-Concept WWTP

50 GPD · Fixed PVA IFAS · CRS Nanobubbles · s::can · AI

Effluent Target: "d10 mg/L NO *f*-N

Site: POC demonstration skid / bench-scale pilot · Generic, US

Document type: Preliminary / Owner Budget Package

Date: July 13, 2026

Process: MLE + Fixed PVA IFAS + CRS NBG + NH₃/NO₃/TSS + AI optimization

Prepared for: Design, permitting, vendor RFQ, and MooreVIEW SCADA integration

Estimates are planning-level (±25–35%) until site-specific engineering and vendor quotes are obtained.

1. Executive Summary

This document defines a proof-of-concept (POC) wastewater treatment plant (WWTP) for POC demonstration skid / bench-scale pilot (Generic, US), rated for 0.035 GPM (50 GPD average flow). The biological process is Modified Ludzack-Ettinger (MLE) biological nitrogen removal configured as hybrid integrated fixed-film activated sludge (IFAS) with fixed PVA foam carriers, CRS (Chemical Reduction Systems) nanobubble generators, s::can recirculating optical measurements (BOD, NH f, NO f, TSS, pH, ORP per phase), and AI-based process optimization.

POC scope: demonstrate "d10 mg/L NO f-N effluent, CRS nanobubble oxygen transfer, and s::can optical analytics with AI-based IR/WAS/DO optimization at 50 GPD (0.035 gpm) before scaling to production. Physical skid: 35 gal anoxic + 65 gal aerobic process tanks (100 gal bio), 270 gal chemical tote (IBC) secondary clarifier, and 200 gal low-profile final tank for polishing and effluent sampling.

Configuration: POC skid — separate process tanks · 270 gal IBC clarifier · 200 gal final. The design target is "d10 mg/L nitrate-nitrogen (NO f-N) in the final effluent, with ammonia-nitrogen (NH f-N) <1 mg/L and secondary quality BOD .../TSS "d10 mg/L.

Metric	Value
Design average flow	50 GPD (0.035 gpm)
Design peak flow	100 GPD
Biological volume	100 gal
Clarifier (IBC tote)	270 gal
Final treatment tank	200 gal
Footprint (approx.)	120-160 ft²
Mid-tier installed CAPEX	~\$165,000
Total project budget (with contingency)	~\$240,000
Annual OPEX (mid)	\$4,000-6,000

2. Design Basis

Parameter	Value	Notes
Average flow (Q)	0.035 GPM (50 GPD)	0.035 gpm design
Peak flow	100 GPD	Diurnal / seasonal peaking
Influent BOD ...	200 mg/L	Carbon for denitrification
Influent BOD ...	200 mg/L	s::can INF_BOD · C:N ratio for denitrification
Influent TSS	180 mg/L	s::can INF_TSS · AI baseline
Influent TKN	40 mg/L	Typical municipal strength
Influent NH f-N	30 mg/L	
Influent total nitrogen	45 mg/L	Includes organic N
Design minimum temperature	10°C	Winter nitrification driver

Effluent NO <i>f</i> -N limit	"d10 mg/L	Monthly average permit target
Effluent NH <i>f</i> -N	<1 mg/L	Design objective
Effluent BOD ... / TSS	"d10 / "d10 mg/L	Secondary treatment standard

3. Process Description

Wastewater passes through preliminary screening (if required by local code), then enters the pre-anoxic zone of the MLE reactor. Return activated sludge (RAS) is mixed with influent in the anoxic zone where denitrification occurs using influent readily biodegradable COD (rbCOD). Mixed liquor underflows to the aerobic IFAS zone where autotrophic nitrifiers oxidize ammonia on both suspended floc and fixed PVA biofilm. Internal recycle (IR) pumps return nitrified mixed liquor from the aerobic zone to the anoxic zone.

Oxygen is supplied primarily through a continuous sidestream loop: mixed liquor is pumped through a CRS Nanobubble Generator (NBG) and returned to the aerobic zone. A small fine-bubble diffuser and trim blower maintains dissolved oxygen (DO) setpoint during peak load or cold weather. Nano-bubbles are not applied in the anoxic zone.

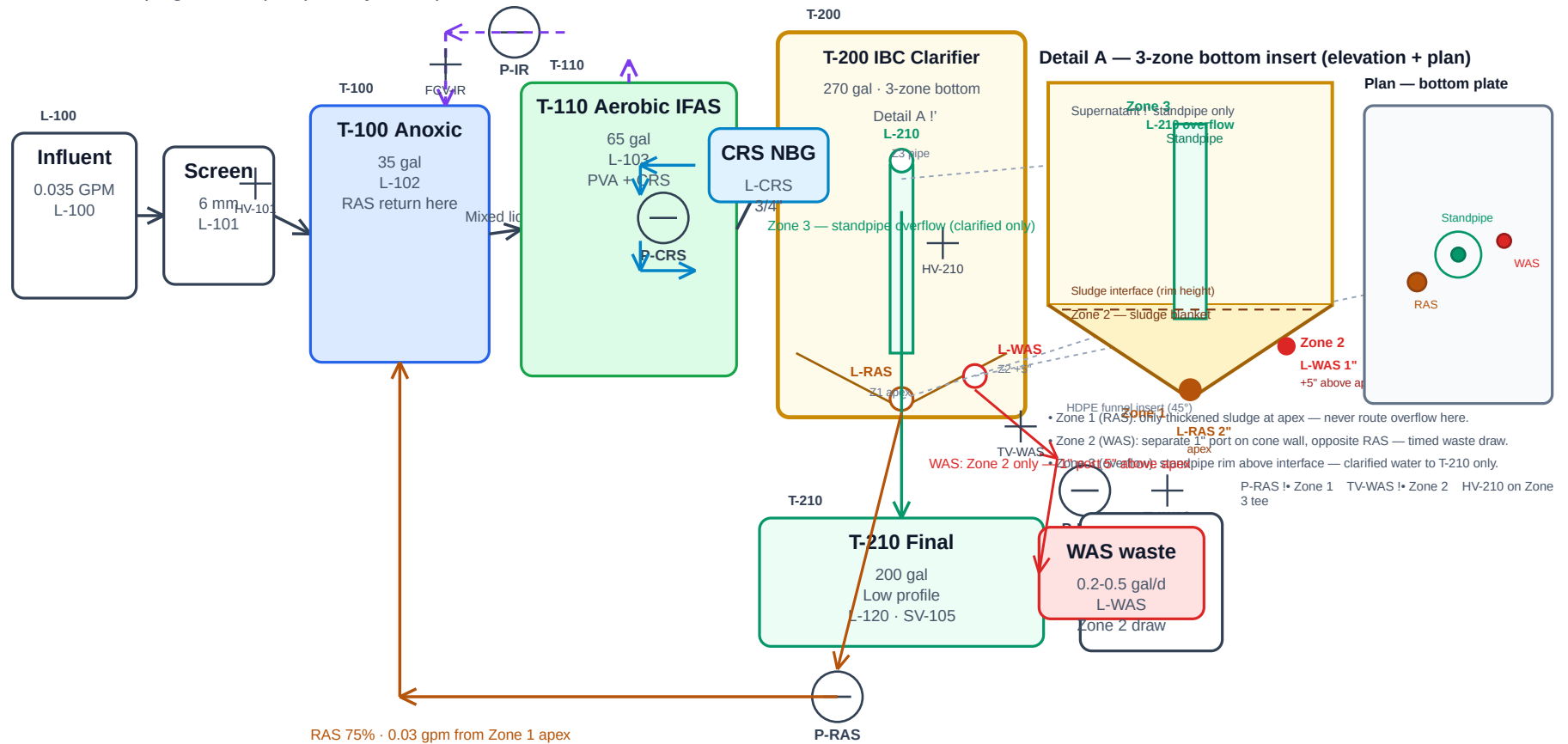
Online measurements use a recirculating s::can optical analyzer package (spectro::lyser, pH::lyser, ORP in flow cell, sam::spector sequencer) that cycles sample ports at each MLE phase: influent (BOD, COD, TSS, NH *f*, pH, UV254), pre-anoxic (NO *f*, ORP, pH, TSS, residual COD), IR stream, aerobic outlet, and final effluent. Influent BOD enables carbon-limited denitrification detection alongside NH *f*/NO *f*/TSS. Aerobic DO remains on dedicated continuous probes for blower PID. All phase tags historize in MooreVIEW; effluent NO *f*-N drives IR VFD trim; AI optimizer uses INF_BOD, C:N ratio, and phase pH/ORP within operator bounds.

POC physical layout: screened influent feeds T-100 (35 gal pre-anoxic) and T-110 (65 gal aerobic IFAS) as separate process tanks on a skid. Mixed liquor overflows to T-200, a 270 gal chemical tote (IBC) used as the secondary clarifier. A 3-zone bottom insert separates hydraulic functions: Zone 1 (RAS sump at cone apex), Zone 2 (WAS draw on cone wall), and Zone 3 (center standpipe overflow for clarified supernatant only via L-210) to T-210, a 200 gal low-profile tank for final hold, polishing, and permit-grade effluent sampling (s::can port SV-105) before discharge.

4. Piping & Instrumentation Diagram (P&ID)

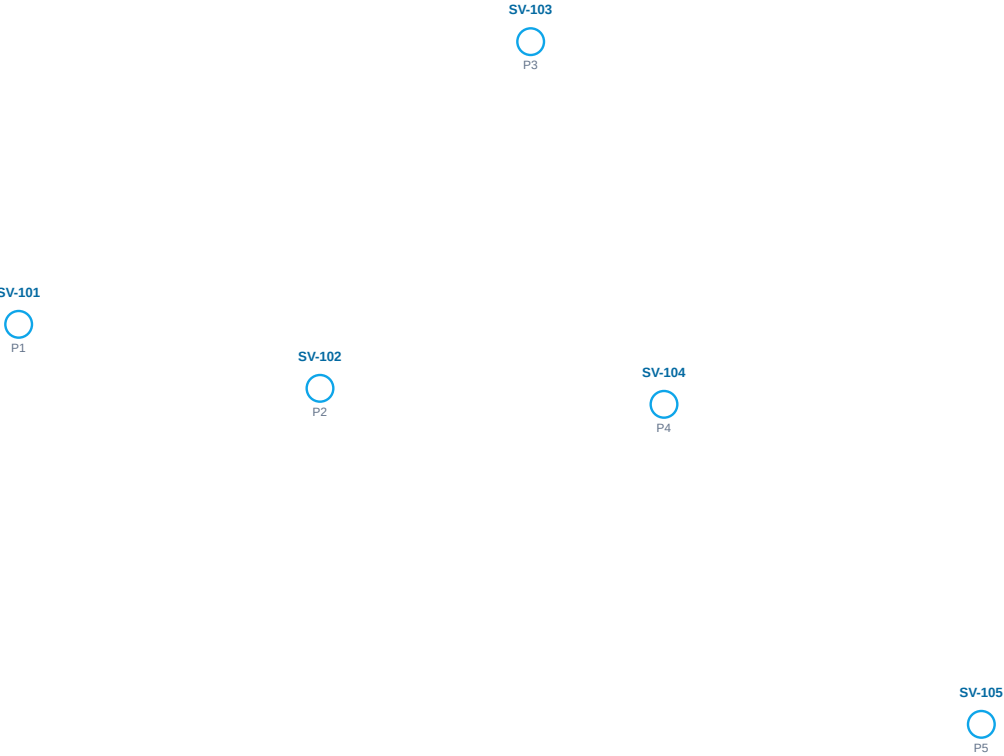
0.035 GPM · Piping, valves, pumps, recycle loops

IR 3.5× Q · 0.12 gpm · L-IR



s::can Package

spectro::lyser V3
pH::lyser · ORP cell
sam::spectro 5-port

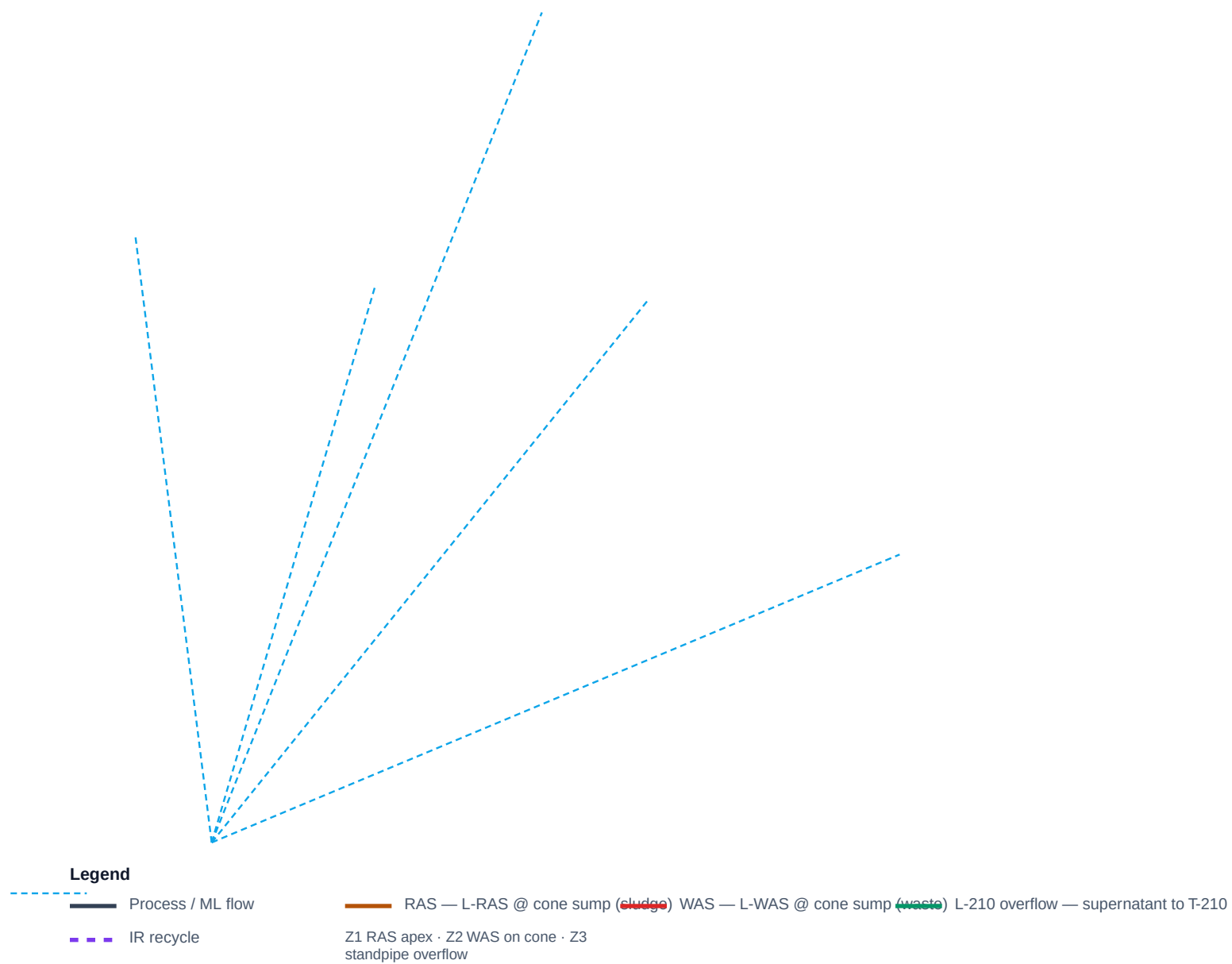


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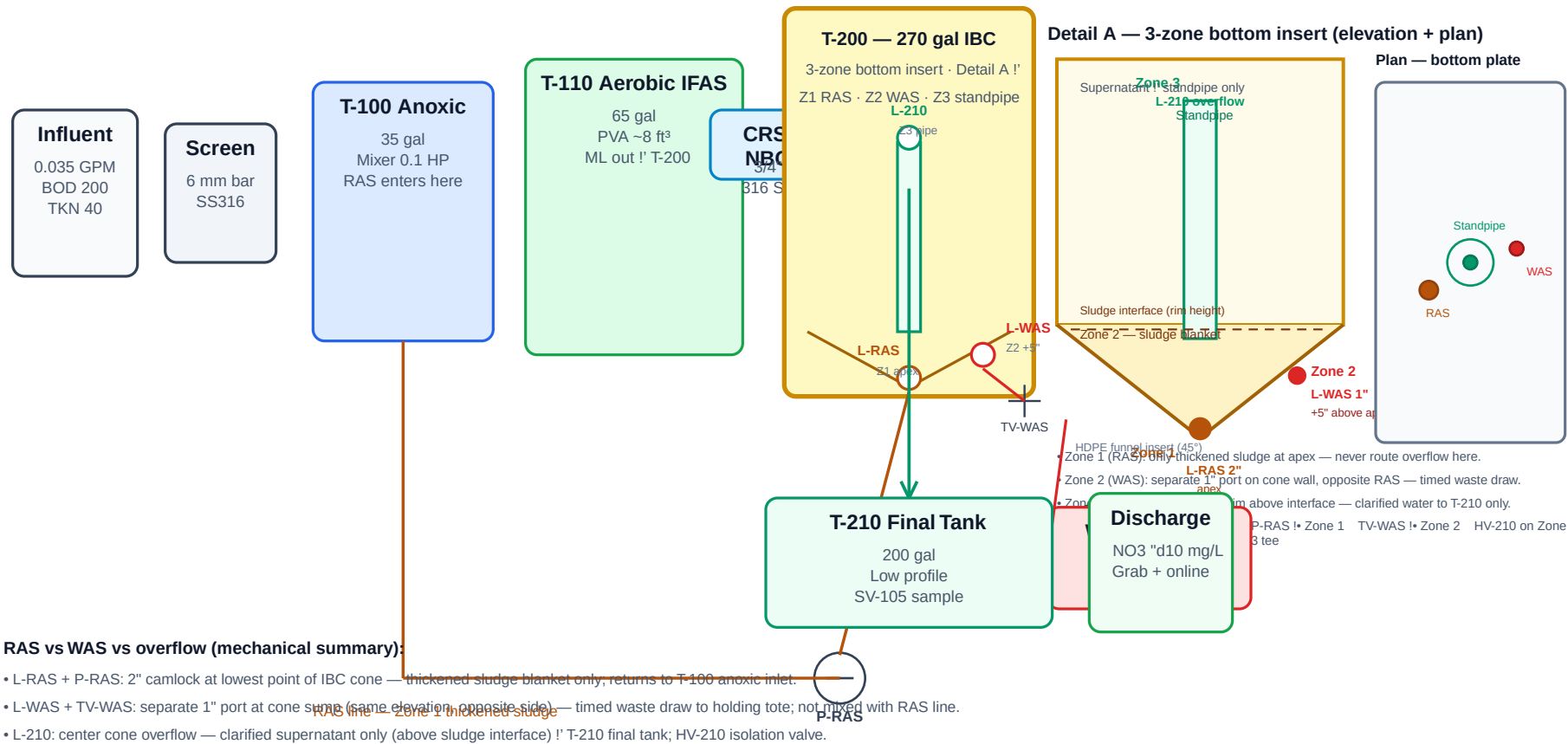
P-SCAN

L-SCAN recirc · return to source



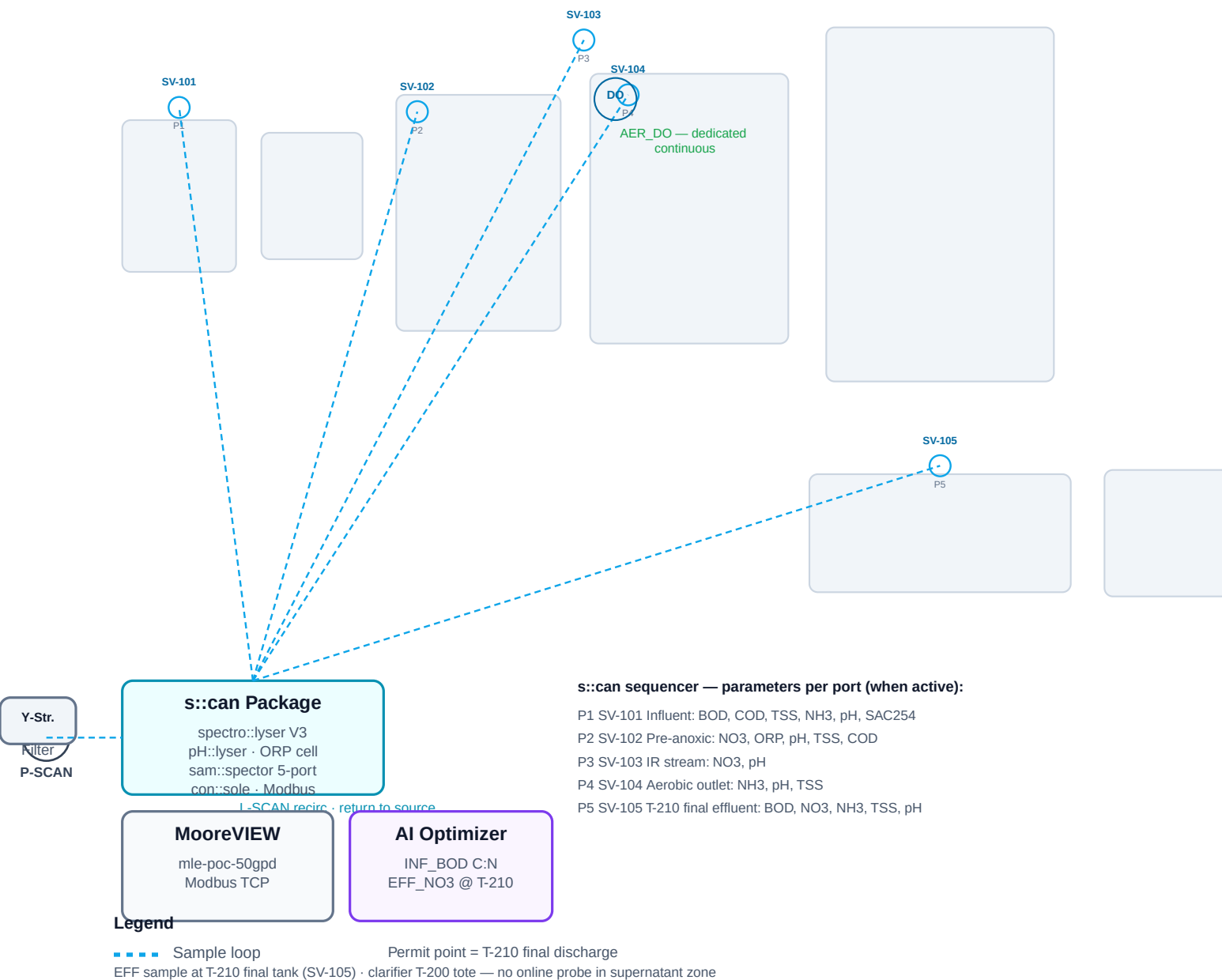
5. Mechanical Drawing

Equipment layout, volumes, materials · POC skid — separate process tanks · 270 gal IBC clarifier · 200 gal final



6. Instrumentation Diagram

s::can recirculating optical loop · per-phase BOD/pH/ORP · MooreVIEW mle-poc-50gpd



7a. Process Flow Summary

Refer to the drawing set on the preceding landscape pages: Sheet 1 (P&ID) includes the s::can recirculating sample loop (L-SCAN, P-SCAN, SV-101...106) to each phase; Sheet 2 (Mechanical) shows the analyzer shelter; Sheet 3 (Instrumentation) lists optical parameters per sequencer port. Main process: influent enters pre-anoxic with RAS; denitrified mixed liquor passes to aerobic IFAS; nitrified liquor is settled and discharged. IR (3.5× Q) returns nitrate-rich liquor to anoxic. CRS sidestream provides nanobubble oxygen in the aerobic zone only.

7b. Process Measurement Points (s::can Recirculating Loop)

Tag	Sample Port	Parameter	Target / Alarm	Control Use
INF_BOD	Influent (SV-101)	BOD	Track ~200 mg/L · C:N	Carbon load · C:N ratio · shock detection
INF_COD	Influent (SV-101)	COD	Historian + AI	Carbon load · C:N ratio · shock detection
INF_TSS	Influent (SV-101)	TSS	Track ~180 mg/L	Carbon load · C:N ratio · shock detection
INF_NH3	Influent (SV-101)	NH3	Track ~30 mg/L	Carbon load · C:N ratio · shock detection
INF_PH	Influent (SV-101)	pH	6.5–8.5	Carbon load · C:N ratio · shock detection
INF_SAC254	Influent (SV-101)	SAC254	UV254 organic load	Carbon load · C:N ratio · shock detection
ANOX_NO3	Pre-anoxic (SV-102)	NO3	<5 mg/L ideal	Denitrification · MLSS · residual carbon
ANOX_ORP	Pre-anoxic (SV-102)	ORP	-50 to +50 mV anoxic	Denitrification · MLSS · residual carbon
ANOX_PH	Pre-anoxic (SV-102)	pH	6.5–8.5	Denitrification · MLSS · residual carbon
ANOX_TSS	Pre-anoxic (SV-102)	TSS	1,500-3,500 mg/L	Denitrification · MLSS · residual carbon
ANOX_COD	Pre-anoxic (SV-102)	COD	Residual carbon proxy	Denitrification · MLSS · residual carbon
IR_NO3	IR stream (SV-103)	NO3	15–30 mg/L typical	IR trim · nitrate in recycle
IR_PH	IR stream (SV-103)	pH	6.5–8.5	IR trim · nitrate in recycle
AER_NH3	Aerobic outlet (SV-104)	NH3	<1 mg/L	Nitrification · plus dedicated AER_DO in basin
AER_PH	Aerobic outlet (SV-104)	pH	6.5–8.5	Nitrification · plus dedicated AER_DO in basin
AER_TSS	Aerobic outlet (SV-104)	TSS	Historian + AI	Nitrification · plus dedicated AER_DO in basin
EFF_BOD	T-210 final effluent (SV-105)	BOD	"d10 mg/L permit	Permit point · T-210 low-profile tank after IBC clarifier
EFF_NO3	T-210 final effluent (SV-105)	NO3	"d10 mg/L permit · IR PID	Permit point · T-210 low-profile tank after IBC clarifier
EFF_NH3	T-210 final effluent (SV-105)	NH3	<1 mg/L	Permit point · T-210 low-profile tank after IBC clarifier
EFF_TSS	T-210 final effluent (SV-105)	TSS	Historian + AI	Permit point · T-210 low-profile tank after IBC clarifier
EFF_PH	T-210 final effluent (SV-105)	pH	6.5–8.5	Permit point · T-210 low-profile tank after IBC clarifier

AER_DO	Aerobic basin (dedicated)	DO	2.0 mg/L setpoint	Not on scan loop — continuous probe
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s::can package: spectro::lyser V3 (UV-Vis BOD, COD, NO₃ f, NH₃ f, TSS, SAC254), pH::lyser in flow cell, ORP at anoxic-phase dwells, con::sole controller, sam::spector 5-port sequencer, P-SCAN recirculation pump with Y-strainer and 10 L buffer. Sample return to source. Dwell 8 min/port; full cycle ~40 min. Calibrate BOD/COD vs lab grab monthly; derive INF_BOD_TKN in MooreVIEW ST. Modbus TCP to edge historian at 1-min scan during active port.

7c. s::can Sequencer Port Schedule

Port	Valve	Phase	Optical Parameters	Notes
1	SV-101	Influent	BOD, COD, TSS, NH3, pH, SAC254	Carbon load · C:N ratio · shock detection
2	SV-102	Pre-anoxic	NO3, ORP, pH, TSS, COD	Denitrification · MLSS · residual carbon
3	SV-103	IR stream	NO3, pH	IR trim · nitrate in recycle
4	SV-104	Aerobic outlet	NH3, pH, TSS	Nitrification · plus dedicated AER_DO in basin
5	SV-105	T-210 final effluent	BOD, NO3, NH3, TSS, pH	Permit point · T-210 low-profile tank after IBC clarifier

7d. AI-Based Process Optimization

The s::can multivariate data stream enables supervised ML that predicts optimal IR ratio, WAS rate, and DO setpoint. INF_BOD and INF_COD quantify carbon available for denitrification; when EFF_NO3 is high but INF_BOD/TKN is low, the model sets AI_CARBON_LIM rather than increasing IR. Phase pH and anoxic ORP provide redox context at each MLE step.

AI Input Tag / Feature	Role in Model
INF_BOD	Influent carbon load; C:N vs TKN; carbon-limit alarm
INF_COD	rbCOD proxy; supplemental carbon decision
INF_BOD_TKN	Derived ratio — denitrification potential
INF_TSS	Solids shock; normalize MLSS
INF_NH3	Nitrogen load forecast
INF_PH	pH inhibition / nitrifier stress
ANOX_NO3 + ANOX_ORP	Denitrification state (single train)
T1_ANOX_* / T2_ANOX_*	Per-train denitrification (dual layout)
ANOX_COD	Residual carbon in anoxic — carbon limited?
IR_NO3	IR effectiveness
AER_NH3 + AER_DO	Nitrification health
EFF_NO3 + EFF_NH3 + EFF_BOD	Permit feedback · model reward
Phase pH (all ports)	Process stability across MLE phases

Recommended AI outputs (constrained setpoints written to MooreVIEW tags):

- AI_IR_SP — suggested IR ratio (bounded 2.5–4.5× Q); suppressed when AI_CARBON_LIM = 1.
- AI_WAS_SP — suggested WAS mL/min from ANOX_TSS trend vs target MLSS band.
- AI_DO_SP — suggested aerobic DO mg/L when AER_NH3 rising and ANOX_TSS stable.
- AI_CARBON_LIM — flag when INF_BOD/TKN < 3.5 or ANOX_COD residual high with ANOX_NO3 > 8 mg/L.
- AI_ALARM — anomaly score when INF_TSS or phase pH deviates >2σ from baseline.

Deployment path: (1) collect 90 days historian data with rule-based IR trim; (2) train regression/gradient model offline on NH₄-N/NO₃-N/TSS features; (3) run inference on edge PC alongside MooreVIEW ST program; (4) validate AI vs manual operation for 30 days before permit reliance. AI never bypasses hard limits on EFF_NO3 >10 mg/L or EFF_NH3 >2 mg/L — those trigger immediate rule-based IR increase and operator alert.

8. Reactor Sizing

Zone	Volume	% of Total	Function
T-100 Pre-anoxic (separate tank)	35 gal	35%	HDPE/poly · mixer only
T-110 Aerobic IFAS (separate tank)	65 gal	65%	PVA + CRS sidestream
T-200 Secondary clarifier (IBC tote)	270 gal	—	3-zone bottom insert · Detail A
T-210 Final treatment / sampling	200 gal	—	Low profile · effluent port SV-105
Total biological	100 gal	100%	HRT "H 2 days @ avg flow

PVA foam media (fixed in SS cages, 5 mm max slot) — total fill ~8 ft³:

- Fill fraction: 50% of aerobic volume (~33 gal displaced)
- Media type: PVA foam cubes/blocks, 10–20 mm
- Surface area: 400–600 m²/m³ (vendor-specific)
- Effective biofilm SRT: 30–60+ days on carriers

9. CRS Nanobubble Aeration System

Equipment: CRS Nanobubble Generator (NBG) by Chemical Reduction Systems (chemicalreduction.com). The CRS NBG is a passive inline/sidestream device using hydrodynamic cavitation and ionization of entrained gases. It requires no electricity, no external gas input, and has no moving parts. Housing: 316 or 304 stainless steel; core alloy suitable for NSF/ANSI 61 applications. Factory warranty: 5 years; design service life: 10+ years.

Component	Specification
CRS NBG model	3/4" line, sidestream per train
Sidestream flow	1-2 continuous
Sidestream pump	0.1 HP, 24/7 duty
Aerobic DO setpoint	2.0 mg/L
Backup aeration	0.05 HP fine-bubble (DO trim)

Anoxic mixing	0.1 HP submersible mixer per train
Estimated O ₂ demand	0.05-0.15 lb O ₂ /d

10. Recycle & Solids Management

Stream	Rate	Equipment
Internal recycle (IR)	3.5× Q = 175 GPD (~0.12 gpm)	VFD centrifugal per train
RAS	75% Q = 38 GPD (~0.03 gpm)	Return to anoxic
WAS	0.2-0.5 gal/d	Timed valve; SRT 12–20 d
MLSS target	2,000-3,500 mg/L	Suspended + biofilm biomass

IR ratio is trimmed on effluent NO₃-N: increase IR when NO₃-N >10 mg/L (max 4.5×); decrease when NO₃-N <6 mg/L (min 2.5×).

10a. T-200 IBC Bottom Insert — Three Hydraulic Zones

Standard flat-bottom IBC totes lack a cone sump and a dedicated overflow weir. For the POC clarifier, install an HDPE funnel insert (HDPE funnel insert, 45° slope, 14" drop to center apex (flat-bottom IBC)) plus a 2" PVC center standpipe, rim adjustable 28–36" above tote floor (sets sludge interface). This creates three physically separated draw points so RAS thickened sludge, WAS blanket waste, and clarified overflow cannot short-circuit. See Detail A on P&ID Sheet 1 and Mechanical Sheet 2.

Zone	Name	Port	Size / elevation	Service
1	RAS sump	L-RAS	2" FNPT / camlock bulkhead · Cone apex — absolute low point	Thickened sludge !' P-RAS !' T-100 anoxic
2	WAS draw	L-WAS	1" FNPT bulkhead + ball valve · Cone slope, 5" vertical above apex (180° from RAS)	Sludge blanket waste !' TV-WAS !' holding tote
3	Overflow (clarified)	L-210	2" tee from standpipe; HV-210 isolation · Standpipe open top / rim — above sludge interface only	Supernatant !' T-210 final tank (not sludge)

- Commissioning: set standpipe rim 2–4" above observed sludge interface after 7–10 days of settle; only supernatant should enter Zone 3.
- Never route overflow through Zone 1 (RAS apex) — that port is for thickened return sludge only.
- WAS (Zone 2) is timed via TV-WAS; draw duration sized for target SRT without draining the RAS sump.

10b. T-200 Bottom Insert — Fabrication BOM & Assembly

Fabricate for 48" L × 40" W (typical 275 gal IBC pallet; verify before fab) IBC. Insert rim 36" × 28" OD at tote floor; 3/8" HDPE (10 mm) machined + welded seams; funnel apex 14" below insert rim plane; WAS port 5" vertical above apex; 180° from RAS on cone slope; standpipe rim 28–36" above tote floor (commission to 2–4" above sludge interface). Standpipe assembly: vertical 2" pipe !' tee branch (L-210) at 12–16" above floor !' union above tee for rim adjustment. Support standpipe to cage; do not hang weight on bulkheads.

Item	Qty	Specification	Install notes
HDPE funnel insert	1	3/8" HDPE sheet; 45° cone; 14" apex drop; rim gasket groove	Machine/weld; bond EPDM rim gasket
RAS bulkhead fitting	1	2" PP/SS tank adapter (Banjo TF200 or equiv.)	Zone 1 — cone apex center; 2" FNPT out
RAS camlock adapter	1	2" male camlock × FNPT	Quick-disconnect for P-RAS suction hose
WAS bulkhead fitting	1	1" PP tank adapter (Banjo TF100 or equiv.)	Zone 2 — drill on cone slope per template
WAS isolation valve	1	1" PVC true-union ball valve (TV-WAS)	Mount outside tote; timed waste draw
Standpipe (vertical)	1	2" PVC Sch 40 pipe, 30" long	Trim to set rim height; solvent-weld to tee
Standpipe tee	1	2" PVC Sch 40 tee (Spears 402-020 or equiv.)	Branch = L-210 overflow; mount 12–16" above floor
Overflow outlet	1	2" PVC Sch 40 × 18" nipple + 90° elbow	Horizontal run through tote wall to L-210
Overflow isolation valve	1	2" PVC true-union ball valve (HV-210)	Downstream of tee branch, outside tote
Standpipe union	1	2" PVC union	Above tee — allows rim height adjustment without cutting
Standpipe support bracket	1	304 SS unistrut or angle + U-bolt	Anchor to tote cage; do not load bulkheads
Insert hold-down	4	1/4" × 2" SS bolt + fender washer + EPDM	Through rim flange into tote floor pad (seal only)
Rim gasket	1	3/16" EPDM strip or closed-cell neoprene	Continuous perimeter under insert rim
PVC cement / primer	1 lot	Weld-On P-68 + 795	Standpipe assembly only; no solvent on HDPE
Wall penetration seal	2	2" and 1" conduit/tank seals (optional)	Where L-210 and L-WAS exit tote sidewall

Assembly sequence:

- Confirm IBC is dedicated clarifier (not chemical service). Drain, rinse, air-dry.
- Template Zone 1 apex, Zone 2 WAS (180° opposite), and standpipe center on tote floor from plan view.
- Install HDPE funnel insert with rim gasket; torque hold-down bolts evenly — watertight, not structural.
- Install 2" RAS bulkhead at apex; 1" WAS bulkhead on cone slope at +5" elevation.
- Solvent-weld standpipe: floor flange or pass-through at center !' tee at 12–16" AFF !' union !' 30" riser (open top).
- Tee branch horizontal through sidewall penetration !' HV-210 !' L-210 to T-210.
- Pressure-test insert rim and bulkheads with water to 6" above standpipe rim before biological startup.
- After 7–10 days operation, adjust standpipe rim 2–4" above observed sludge interface.

11. Expected Performance

Parameter	Expected Effluent
BOD ...	5–8 mg/L

TSS	5–10 mg/L
NH <i>f</i> -N	<1 mg/L
NO <i>f</i> -N	6–10 mg/L (with IR trim)
Total nitrogen (TN)	10–14 mg/L

Nitrogen balance: ~35–40 mg/L NO *f*-N formed after full nitrification must be reduced ~75–80% via MLE denitrification to achieve <10 mg/L NO *f*-N. At BOD 200 mg/L and TKN 40 mg/L, influent rbCOD (~100–120 mg/L) is adequate without supplemental carbon under normal conditions.

12. Equipment Schedule

Item	Qty	Size / Model
T-100 Anoxic process tank	1	35 gal HDPE/poly
T-110 Aerobic IFAS process tank	1	65 gal HDPE/poly
PVA foam + retention cages	1 lot	~8 ft ³ ; 50% aerobic fill
T-200 Secondary clarifier	1	270 gal chemical tote (IBC)
T-200 3-zone bottom insert kit	1	HDPE funnel + RAS/WAS bulkheads + standpipe/tee — \$10b BOM
T-210 Final / sampling tank	1	200 gal low-profile
CRS NBG sidestream unit	1	3/4" SS; RFQ from CRS
Sidestream recirculation pump	1	1-2; 0.1 HP
Backup fine-bubble blower	1	0.05 HP with DO control
Anoxic submersible mixer	1	0.1 HP
IR pump (VFD)	1	0.12 gpm @ 8 ft TDH
RAS pump (tote sludge)	1	0.03 gpm
s::can recirculating analyzer package	1	s::can · 5-port loop · BOD/pH/ORP
AER_DO probe (dedicated)	1	Continuous · blower PID — not on scan loop
MooreVIEW edge SCADA + AI module	1	Project: mle-poc-50gpd

s::can installed budget ~\$55k (included in Section 16 CAPEX). Monthly lab verification of spectro::lyser BOD/COD vs Standard Methods required for permit-grade reporting.

13. Instrumentation & Control

Tag	Sample Port	Function
INF_BOD	Influent (SV-101)	Carbon load · C:N ratio · shock detection
INF_COD	Influent (SV-101)	Carbon load · C:N ratio · shock detection
INF_TSS	Influent (SV-101)	Carbon load · C:N ratio · shock detection
INF_NH3	Influent (SV-101)	Carbon load · C:N ratio · shock detection

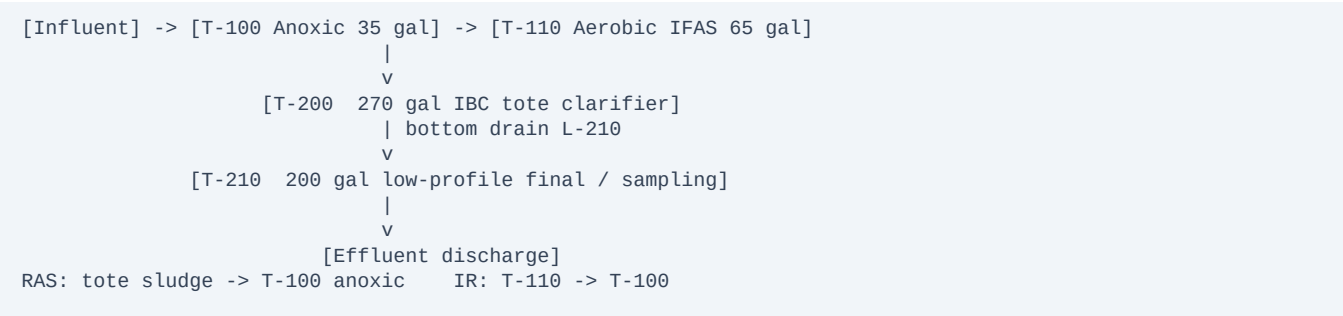
INF_PH	Influent (SV-101)	Carbon load · C:N ratio · shock detection
INF_SAC254	Influent (SV-101)	Carbon load · C:N ratio · shock detection
ANOX_NO3	Pre-anoxic (SV-102)	Denitrification · MLSS · residual carbon
ANOX_ORP	Pre-anoxic (SV-102)	Denitrification · MLSS · residual carbon
ANOX_PH	Pre-anoxic (SV-102)	Denitrification · MLSS · residual carbon
ANOX_TSS	Pre-anoxic (SV-102)	Denitrification · MLSS · residual carbon
ANOX_COD	Pre-anoxic (SV-102)	Denitrification · MLSS · residual carbon
IR_NO3	IR stream (SV-103)	IR trim · nitrate in recycle
IR_PH	IR stream (SV-103)	IR trim · nitrate in recycle
AER_NH3	Aerobic outlet (SV-104)	Nitrification · plus dedicated AER_DO in basin
AER_PH	Aerobic outlet (SV-104)	Nitrification · plus dedicated AER_DO in basin
AER_TSS	Aerobic outlet (SV-104)	Nitrification · plus dedicated AER_DO in basin
EFF_BOD	T-210 final effluent (SV-105)	Permit point · T-210 low-profile tank after IBC clarifier
EFF_NO3	T-210 final effluent (SV-105)	Permit point · T-210 low-profile tank after IBC clarifier
EFF_NH3	T-210 final effluent (SV-105)	Permit point · T-210 low-profile tank after IBC clarifier
EFF_TSS	T-210 final effluent (SV-105)	Permit point · T-210 low-profile tank after IBC clarifier
EFF_PH	T-210 final effluent (SV-105)	Permit point · T-210 low-profile tank after IBC clarifier
AER_DO	Aerobic basin (dedicated)	Not on scan loop — continuous probe
SCAN_ACTIVE_PORT	Sequencer	Current port 1–N
SCAN_RUN / SCAN_FAULT	P-SCAN loop	Pump status / system fault
AER_DO	Aerobic basin	Continuous DO 2.0 mg/L; blower PID
AI_IR_SP	AI Optimizer out	Suggested IR setpoint (Auto mode)
AI_WAS_SP	AI Optimizer out	Suggested WAS rate from TSS trend
AI_CARBON_LIM	AI Optimizer out	Carbon-limited denitrification flag
IR_FLOW / IR_RATIO	IR pump	Flow and ratio monitoring
TANK_LVL	Reactor	Level alarm
NANO_LOOP_RUN	Sidestream pump	CRS loop runtime / fault

Control logic summary:

- Sidestream CRS loop: run continuous (24/7).
- s::can sequencer: run continuous; alarm SCAN_FAULT; log all ports each cycle.
- IF EFF_NO3 > 10 mg/L AND NOT AI_CARBON_LIM !' increase IR +0.2×/hr (max 4.5×).
- IF AI_CARBON_LIM = 1 !' hold IR; alarm operator; evaluate supplemental carbon.

- IF EFF_NO3 < 6 mg/L !' decrease IR " 0.2×/hr (min 2.5×).
- IF AER_NH3 > 2 mg/L OR EFF_NH3 > 1 mg/L !' increase DO or reduce WAS.
- IF ANOX_TSS > 4,000 mg/L for 4 hr !' increase WAS per AI_WAS_SP; check settler.
- IF INF_BOD/TKN < 3.5 (derived tag) !' set AI CARBON LIM; do not raise IR.
- IF ANOX_ORP > +100 mV at anoxic port dwell !' alarm: denitrification lost.
- IF phase pH < 6.2 or > 8.8 at any port !' process pH alarm.

14. Tank Layout (Section)



15. Startup & Commissioning

- Fill reactor; seed with nitrifying return sludge or commercial culture.
- Install and cage PVA media permanently — floating media causes clarifier carryover.
- Commission anoxic mixer and IR/RAS loops before enabling CRS sidestream.
- Start CRS sidestream at 50% pump speed; ramp while monitoring AER_DO and ANOX_ORP.
- POC: operate at 50 GPD steady flow for 14 days before step-load tests; log all s::can tags at 1-min scan.
- Tune IR trim after 30 days of stable operation and seasonal temperature data.

16. Capital Cost Estimate (CAPEX)

Planning-level installed costs, USD, 2025–2026. ±25–35% until vendor quotes received.

Category	Budget	Mid	High
MLE reactor tanks (1×125K or 100 gal)	\$21k	\$36k	\$64k
Fixed PVA foam + cages	\$5k	\$13k	\$26k
CRS sidestream aeration (all trains)	\$6k	\$12k	\$23k
Pumps, mixers, clarifiers	\$13k	\$25k	\$46k
Instrumentation + AI	\$9k	\$20k	\$38k
Controls / SCADA	\$5k	\$10k	\$20k
Electrical + piping + site	\$16k	\$30k	\$56k
Install + commissioning	\$11k	\$20k	\$36k
Engineering + FDEP permits	\$9k	\$17k	\$31k
TOTAL INSTALLED	\$107k	\$165k	\$256k

17. Project Budget Summary

Line Item	Amount
Construction (mid installed)	~\$165,000
Owner contingency (15%)	~\$24,750
Collection / EQ / site civil (allowance)	~\$24,750
First-year OPEX reserve	~\$3,300
TOTAL PROJECT BUDGET	~\$240,000

Excluded from above: land acquisition, long collection systems, enclosed buildings, effluent UV/chlorination, and septage receiving.

18. Operating Cost Estimate (OPEX)

Item	Annual Cost
Power (CRS loop + trim blower)	\$120 – \$180
Lab / compliance sampling	\$800 – \$2,500
Analyzer consumables	\$0 – \$1,500
WAS hauling	\$200 – \$600
PVA media replacement reserve	\$300 – \$800
Maintenance contract	\$500 – \$2,000
Contract O&M visits	\$1,000 – \$3,000
TOTAL ANNUAL OPEX	\$4,000-6,000

19. CRS Vendor RFQ Template

To: Chemical Reduction Systems (CRS) – chemicalreduction.com
Subject: RFQ – CRS NBG for 50 GPD MLE-IFAS WWTP (POC demonstration skid / bench-scale pilot)

Application: 50 GPD MLE-IFAS, POC skid – separate process tanks · 270 gal IBC clarifier · 200 gal final, fixed PVA biofilm carriers, effluent target <=10 mg/L NO₃-N.

Request: 1× CRS NBG sidestream unit(s), 3/4" line, 316 SS,
1-2 continuous recirculation on mixed liquor per train.

Please quote: unit price, min/max flow, pressure drop, lead time, warranty,
WWTP reference installations, and recommendation for backup aeration at 10°C.

20. Permitting & Design Notes

- POC objective: validate MLE + PVA IFAS + CRS nanobubbles + scan optical loop + AI control at 50 GPD before full-scale deployment.
- Physical layout: individual 35 gal anoxic + 65 gal aerobic process tanks; 270 gal chemical tote (IBC) as secondary clarifier with 3-zone bottom insert (RAS sump / WAS draw / standpipe overflow); bottom drain to 200 gal low-profile final tank.

- Biological HRT ~2 days on 100 gal process volume @ 50 GPD; clarifier and final tanks provide surge capacity and permit-grade sample point.
- Permit: may operate under pilot/research authorization or temporary permit; confirm with local authority.
- S::can recirculating loop and MooreVIEW historian capture 90+ days training data for AI model.
- If limit is NO3 ion at 10 mg/L (~2.3 mg/L as N), add post-denitrification filter.

21. Document Revision

Rev	Date	Description
0	2026-07-13	Initial preliminary design package